

Team name: Fusion Zero

Currency: AUD and USD. USD values use an estimated conversion rate of 1 AUD = 0.70 USD. Costs are approximate and listed for one competition robot.

Hardware total: AUD \$795.86 / USD \$557.10. Software total: AUD \$0.00 / USD \$0.00.

Hardware

#	Component	Part name	Author / Manufacturer	Source	Qty	Unit AUD	Total AUD	Unit USD	Total USD	Notes
1	Distance sensor	VL53L0X Time of Flight sensor module	STMicroelectronics / module seller	AliExpress	3	\$5.00	\$15.00	\$3.50	\$10.50	Front, left and right distance sensing
2	IMU	BNO085 / BNO08x 9-axis IMU module	CEVA / Bosch / module seller	AliExpress	1	\$35.00	\$35.00	\$24.50	\$24.50	Pitch, roll and yaw feedback
3	Battery holder	3S plated 18650 battery pack holder	Module seller	AliExpress	1	\$3.00	\$3.00	\$2.10	\$2.10	Holds 3 x 18650 cells in series
4	Switch	Small SPDT switch	Module seller	AliExpress	1	\$1.00	\$1.00	\$0.70	\$0.70	Mode / auxiliary switching
5	Power switch	Larger SPDT power switch	Module seller	AliExpress	1	\$1.00	\$1.00	\$0.70	\$0.70	Main robot power switch
6	Touch sensors	SPDT micro switches	Module seller	AliExpress	4	\$1.00	\$4.00	\$0.70	\$2.80	Collision and wall contact detection
7	Servo	SG90 micro servo	TowerPro / module seller	AliExpress	1	\$5.00	\$5.00	\$3.50	\$3.50	Auxiliary mechanism actuation
8	Servo	DSSERVO RDS3218 270 degree 20 kg servo	DSSERVO	AliExpress	1	\$30.00	\$30.00	\$21.00	\$21.00	Main rescue claw actuation
9	Storage	128 GB microSD card	Various	AliExpress	1	\$30.00	\$30.00	\$21.00	\$21.00	OS and robot code storage
10	Lighting	5 V COB LED light strip	Module seller	AliExpress	1	\$5.00	\$5.00	\$3.50	\$3.50	Consistent lighting for vision
11	Battery cells	Samsung INR18650-35E 3500 mAh 10 A cells	Samsung SDI	TinkerTech AU	3	\$12.00	\$36.00	\$8.40	\$25.20	Main robot power cells
12	Drive motors	380:1 Micro Metal Gearmotor HPCB 6 V	Pololu	Pololu product 4796	4	\$37.79	\$151.16	\$26.45	\$105.81	Four drive motors
13	Wheel hubs	Universal Aluminum Mounting Hub for 4 mm Shaft, M3 holes, 2-pack	Pololu	Pololu product 1997	2	\$15.64	\$31.28	\$10.95	\$21.90	Two packs provide four hubs
14	Motor driver	TB6612FNG Dual Motor Driver Carrier	Pololu	Pololu product 713	2	\$7.07	\$14.14	\$4.95	\$9.90	Drives the four DC gearmotors
15	Voltage regulator	5 V, 6 A Step-Down Voltage Regulator D42V55F5	Pololu	Pololu product 5571	2	\$45.64	\$91.28	\$31.95	\$63.90	Regulated 5 V rails
16	Controller	Raspberry Pi 5 4 GB	Raspberry Pi Ltd	Raspberry Pi	1	\$109.00	\$109.00	\$76.30	\$76.30	Main computer for vision and logic
17	Camera	Raspberry Pi Camera Module 3 Wide	Raspberry Pi Ltd / Sony	Raspberry Pi	1	\$62.00	\$62.00	\$43.40	\$43.40	Downward line and colour vision
18	Camera	Raspberry Pi AI Camera	Raspberry Pi Ltd / Sony	Raspberry Pi	1	\$112.00	\$112.00	\$78.40	\$78.40	Evacuation vision and AI capable camera
19	Custom PCB	Fusion Zero custom PCB and assembled parts	Fusion Zero / JLCPCB	JLCPCB	1	\$30.00	\$30.00	\$21.00	\$21.00	JST/XH headers, MOSFET, capacitors and switches; sponsored
20	Microcontroller	Seeed Studio XIAO ESP32C3	Seeed Studio	Seeed Studio	1	\$10.00	\$10.00	\$7.00	\$7.00	Auxiliary microcontroller
21	Mechanical parts	3D printed parts, screws, wires, standoffs and small fixings	Fusion Zero / various	Team fabricated / various	1	\$20.00	\$20.00	\$14.00	\$14.00	Chassis, mounts, claw parts and hardware

Software and Tools

#	Component	Part name	Author / Manufacturer	Source	Qty	Unit AUD	Total AUD	Unit USD	Total USD	Notes
1	Operating system	Raspberry Pi OS	Raspberry Pi Ltd	Raspberry Pi	1	\$0.00	\$0.00	\$0.00	\$0.00	Robot runtime environment
2	Programming language	Python 3	Python Software Foundation	python.org	1	\$0.00	\$0.00	\$0.00	\$0.00	Main robot software language
3	Computer vision	OpenCV	OpenCV project	opencv.org	1	\$0.00	\$0.00	\$0.00	\$0.00	Line detection, colour masking and contours
4	Camera library	Picamera2	Raspberry Pi Ltd	Raspberry Pi	1	\$0.00	\$0.00	\$0.00	\$0.00	Camera capture and configuration
5	GPIO interface	gpiozero / I2C / Raspberry Pi GPIO stack	Raspberry Pi community	Raspberry Pi / Python packages	1	\$0.00	\$0.00	\$0.00	\$0.00	Digital inputs, switches and auxiliary control
6	I2C framework	Adafruit Blinka and CircuitPython libraries	Adafruit	Adafruit / PyPI	1	\$0.00	\$0.00	\$0.00	\$0.00	Hardware abstraction for I2C sensors
7	Distance sensor library	Adafruit CircuitPython VL53L0X	Adafruit	Adafruit / PyPI	1	\$0.00	\$0.00	\$0.00	\$0.00	Reads VL53L0X sensors
8	IMU library	Adafruit CircuitPython BNO08x	Adafruit	Adafruit / PyPI	1	\$0.00	\$0.00	\$0.00	\$0.00	Reads BNO085 orientation data
9	Development environment	Visual Studio Code with Remote SSH	Microsoft	Microsoft	1	\$0.00	\$0.00	\$0.00	\$0.00	Remote coding and debugging
10	Version control	Git and GitHub	Git / GitHub	github.com	1	\$0.00	\$0.00	\$0.00	\$0.00	Source control and collaboration
11	CAD	Onshape	PTC	onshape.com	1	\$0.00	\$0.00	\$0.00	\$0.00	Mechanical design and 3D printed parts
12	PCB design	EasyEDA	EasyEDA	easyeda.com	1	\$0.00	\$0.00	\$0.00	\$0.00	Custom PCB design
13	Robot code	Fusion Zero custom robot software	Fusion Zero	Team developed	1	\$0.00	\$0.00	\$0.00	\$0.00	Line following, evacuation, sensor fusion and state machine

Grand total AUD	\$795.86
Grand total USD	\$557.10

Notes: JLCPCB fabrication was sponsored, but the estimated fabrication and assembled parts cost is shown for BOM transparency. Small parts such as connectors, wiring, fixings and printed parts are grouped as permitted by the BOM format.